

Subject: CGDate: 9/9/24

• Sec-A

1. What is Computer Graphics?

Ans: Computer Graphics is the field of computer science that deals with creating, manipulating, and displaying visual content, such as images, animations, and simulations using computers. It involves techniques and produce interactive graphical interfaces for various applications like videogames, simulations & design software.

2. What is Resolution?

Ans: Resolution refers to the amount of detail an image holds, typically measured by the number of pixels in an image or displays. Higher Resolution means more pixels and finer detail.

3. What is Persistence?

Ans: Persistence refers to how long a phosphor dot on a screen continues to glow after the electron beam has moved away. It is the duration a visual image remains visible on a display before fading.

4. What is Aspect Ratio?

Ans: Aspect ratio is the proportional relationship between the width and height of an image or screen, typically expressed as width.

ex: 16:9

$$\text{Aspect Ratio} = \text{Width} / \text{Height}$$

ex: 16:9, in a 16:9 aspect ratio, for every 16 units of width, there are 9 units of heights.

5. What is horizontal & vertical retrace of the electron beam?

Ans Horizontal retrace is when the electron beam moves from the right to the left side of the screen after drawing a line.

Vertical retrace is when the beam moves from the bottom to the top of the screen after drawing all lines to start the next frame.

• Sec-B

1. Differentiate betⁿ Random scan and Raster scan display.

Ans Random scan	Raster scan
(i) it has high Resolution.	its Resolution is low.
(ii) it is more expensive.	2. it is less expensive.
(iii) Any modification if needed is easy.	3. Modification is tough.
(iv) Solid pattern is tough to fill.	4. Solid pattern is easy to fill.
v. Refresh rate depends on resolution.	5. Refresh rate does not depend on the picture.
vi. in vector scan display, the beam is moved between the end points of the graphics primitives.	6. in raster scan display the beam is moved all over the screen one scan line at a time, from top to bottom & then back to top.
(vii). Vector/Random display only draws line and characters.	7. Raster display has ability to display areas filled with solid colours or patterns.
(viii) Only screen with view on an area is displayed.	8. Whole screen is scanned.

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2. Explain Beam-penetration & Shadow-mask methods.

Beam-penetration and shadow-mask are two techniques used for producing color images on CRT (cathode ray tube) displays.

(i) Beam-penetration Method :-

How it works - This method uses two layers of phosphor, typically red and green, coated on the CRT screen. The electron beam excites the phosphors to produce different colors.

• Color Production -

(a) Low-energy electrons excite the outer green phosphor layer, producing green light.

(b) High-energy electrons excite the outer green phosphor layer, producing green light.

(c) High energy electrons penetrate deeper, exciting the inner red phosphor layer, producing red light.

(d) By varying the beam intensity, intermediate colors like yellow or orange can be created through the combination of red and green.

• Advantage: Simple and inexpensive to implement.

• Disadvantage - Limited to only a few colors (red, green, yellow, orange).

• Uses - Mainly used in early color displays.

(ii) Shadow-mask Method:

How it works - This method uses three electron guns (for red, green & blue) and a perforated metal mask (the shadow mask) positioned just behind the screen. Each gun directs electrons towards its respective color phosphor dots on the screen (red, green, or blue).

• Color production -

(a) The shadow mask ensures that each electron beam only hits the correct color phosphor dots.

(b) By varying the intensity of each gun, different combinations

of red, green, and blue light can be mixed to produce a wide range of colors.

(c) The combination of these three primary colors (RGB) can create millions of colors.

- Advantages:- (a) Capable of producing high-quality, full color images.
- (b) More accurate color reproduction with a wider color range.

- Disadvantage - More expensive & complex to manufacture.

- Use - Used in most modern CRT color television and computer monitors.

Q3. Explain some input device which used in computer graphics.

Ans input devices in CG are used to interact with and manipulate graphic data. Here are some common input devices -

(i) Keyboard -

- Function - Standard input device used for typing text and providing commands.

- Usage in graphics - Often used for shortcuts, commands, and entering values, especially in design software or CAD systems.

(ii) Mouse:

- Function - A pointing device that detects two-dimensional motion relative to a surface.

- Usage in graphics - Used for selecting, dragging, and manipulating objects in GUI and for drawing in design applications.

(iii) Trackball:

- Function - A pointing device that consists of a ball held in a socket and rotated manually.

(iv) Light Pen:-

- Function - A light sensitive pen like device that allows users to point to or draw directly on the display screen.

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- Uses in graphics - Primarily used in early CAD system and G2.

(v) Joystick -

- Function - A device that allows user to control objects by tilting a lever in different dirⁿ.
- Usage in Graphics - Used in some gaming or 3D modeling application for rotating objects.

(vi) Scanner -

- Function - Convert physical images or documents into a digital form.
- Usage in graphics - Used to digitize photos, drawings, or other physical media.

(vii) Touch Screen -

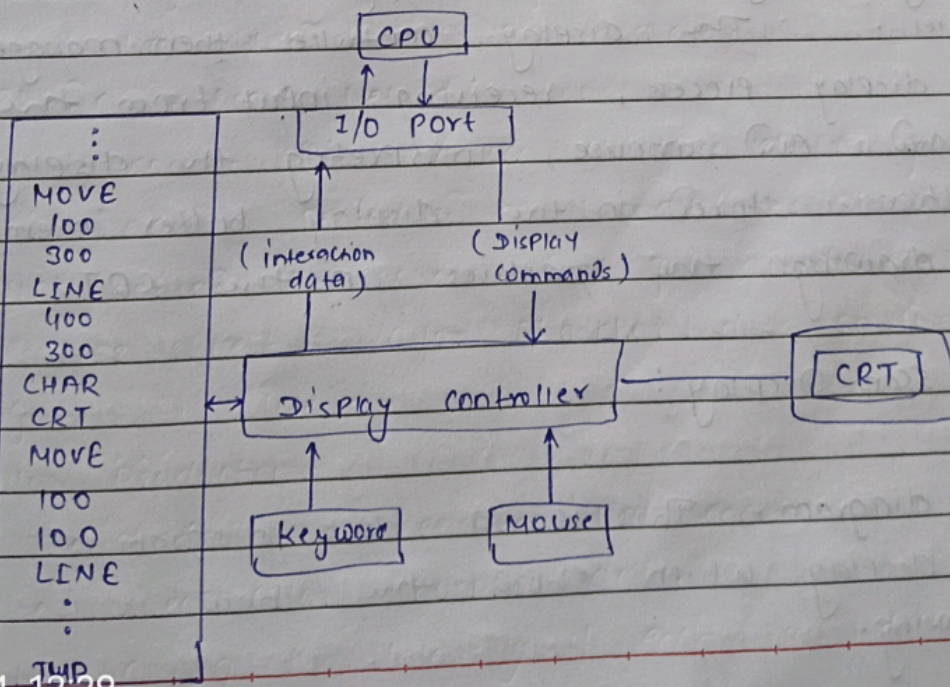
- Fun. - A display that also serves as an input device, responding to touch gestures.

• Sec-C

1. Sketch the block diagram of random and raster scan display and explain in details.

Ans (i) Random scan Display (vector Display) :

Block Diagram - This is a arch. diagram which represents how Raster scan display system works.



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Components -

- CPU - The Central Processing unit is responsible for overall system control and processing data, including the display data.
- I/O port - This is the interface betⁿ the CPU and the external devices, including the display controller.
- Display controller - This component is responsible for managing the display process. It receives display commands from the CPU and interaction data from input devices like the keyboard and mouse.
- Keyboard and mouse - These are input devices used to interact with the computer and provide input data to the display controller.
- CRT - The Cathode Ray Tube is the display device that produces the visual output.
- Display buffer memory - This is a memory area where the display data is stored.

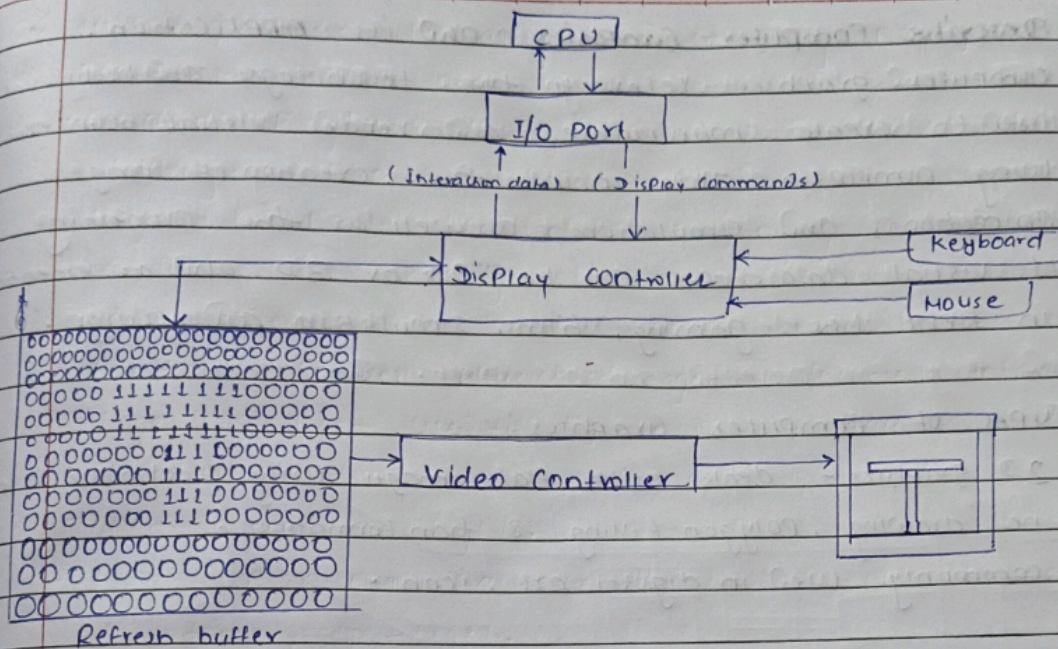
Working - The random scan display arch. involves the CPU processing display data and sending it to the display controller. The display controller then manages the display process, receiving input from the keyboard and mouse, interpreting the display instructions stored in the display buffer memory, and drawing the graphics on the CRT.

(ii) Raster Scan Display :-

Block diagram - This is a block diagram of raster scan display which shows how it is working.

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- **Display controller** - This component is responsible for managing the display process.
- **Video Controller** - This component generates the video signal that is sent to the display device (ex- a monitor) to produce the visual output.
- **Refresh buffer** - This is a memory area where the display data is stored. it is continuously refreshed to maintain the displayed image.

The Raster Scan display arch. involves the CPU processing display data and sending it to the display controller. The display controller then manages the display process, receiving input from the keyboard and mouse, generating the video signal, and refreshing the display buffer to maintain the image on the screen.

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Q2 Describe Computer Graphics and its Application.

Ans Computer graphics refers to the technology and techniques used to create, manipulate, and render visual content using computers. This includes the creation of images, animations, and simulations, as well as the processing of visual data. CG can be 2D or 3D and is essential in fields like gaming, film, simulation and design.

- Types of Computer Graphics:-

(i) 2D graphics - deals with flat images using techniques like line drawing, polygon filling & transformations.

→ commonly used in digital art, icon.

(ii) 3D Graphics - involves the creation of objects with height, width and depth. Commonly used in movies, video games and CAD app.

(iii) Image Processing - Modifies or manipulates images to enhance or analyze them.

(iv) Animation - Involves creating moving images using techniques like frame-by-frame animation or motion interpolation.

- Application of Computer Graphics:-

(i) Entertainment and media -

- Movies and Animation - Computer graphics are used to create visual effects (VFX) and animations in films.

- Video Games - The gaming industry uses advanced 3D graphics for real-time rendering of complex virtual worlds and characters.

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Interact with a fully virtual world, while AR overlays digital images onto the real world.

(iii) Computer-Aided Design (CAD): -

- Engineering and Architecture - CAD applications rely on computer graphics to design everything from mechanical parts to entire buildings.
- Product design - In manufacturing, CAD are used to design and test products virtually, reducing the need for physical prototypes.

(iii) Education and Training -

- Educational SW - Interactive learning tools often incorporate computer graphics to make complex concepts more engaging.

(iv) Science & Research -

- Visualization - Scientists use computer graphics to visualize complex data sets, such as medical images, climate models, and molecular structures.
- Simulation - Computer simulations can be used to model scientific phenomena that would be difficult or impossible to study in the real world.

(v) Business and Marketing -

- Presentation - CW can enhance presentation with visually appealing charts, graphs, and diagrams.
- Advertising - Eye-catching visuals created using computer graphics are essential for advertising campaigns.

(vi) Art & Design -

- Digital Art - Artists use graphic SW like Adobe Photoshop or Corel Painter to create digital illustrations, painting and design.

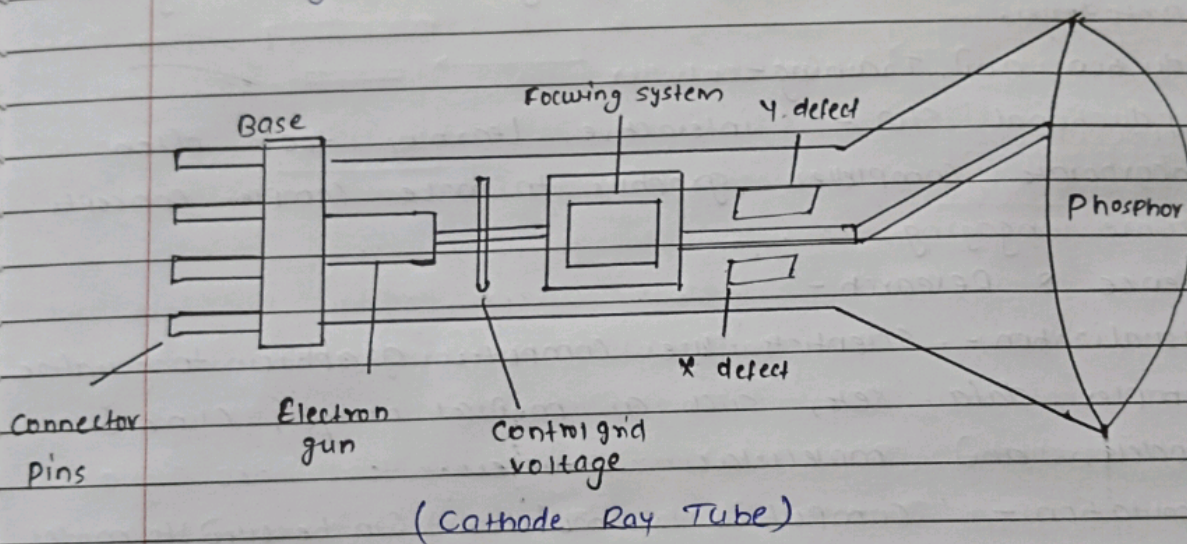
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Q3 Explain CRT monitor with diagram.

Ans CRT stands for Cathode Ray tube. CRT is a Technology used in traditional computer monitors and televisions. The image on CRT display is created by firing electrons from the back of the tube at Phosphorus located towards the front of the screen.

Once the electron heats the phosphorus, they light up, and they are projected on a screen. ~~The colour~~



(Cathode Ray Tube)

Components of CRT monitor :-

- (i) Electron gun - Generates and emits electron beam towards the screen. In color CRTs, there are three guns, each for red, green, and blue colors.
- (ii) Focusing System - This system ensures the electron beams are sharply focused to a point when they strike the screen.
- (iii) Deflection System - Control the direction of the electron beams to scan across the screen. It consists of horizontal and vertical deflection coils that guide the beams in raster lines.
- (iv) Phosphor Coating - A layer of Phosphor dots on the screen's inner surface. These Phosphors glow when

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Struck by the electron beams, Producing visible light.

(V) Phosphor-coated screen- The front glass face of the CRT, where the phosphor dots glow to form images when hit by the electron beams.

• Working Principle:-

- The electron gun generates and fires electrons towards the phosphor-coated screen.
- The deflection system controls the electron beam's movement, ensuring it scans horizontally and vertically in a raster pattern.
- The beam hits the phosphor coating, causing it to glow at the hit points, forming an image on the screen.
- In color CRTs, three electron beams (red, green, blue) hit their respective phosphor dots through the shadow mask, creating a colored image.

Altogether, through CRT monitors have been replaced by more efficient and compact display technologies. They were a crucial development in the history of display technology, offering a foundation for the evolution of modern screens.

The end.